Guideline

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A. Basic usage

1. Prepare initial model files. (*POSCAR_init*)

   The POSCAR flavor of *VASP*.

2. Prepare the control file for *MyTm*. (*input.dat*)
   Generate a pre-configured *input.dat* file using the following command:
   MyTm -m para

   para is the integer number representing the melting point calculation method. (0: Direct-heating method; 1: void method; 2: modified void method;3: two-phase method; 4: sandwich method; 5: Z method; 6: modified Z method;)

3. Prepare MD call script. (see section C)
4. Melting point calculation

   Start melting point calculation using the following command:

   MyTm -r

B. Explanation of all commands in the *input.dat* file

The command is divided into four parts, controlling different modules for melting point calculation. These four parts are solid-liquid atomic classification, ensemble selection (MD call), temperature control, and model construction.

1. solid-liquid atomic classification

Name	Default	Description
method	Command (None)	The method used by *MyTM* for solid-liquid atomic classification ***1*** : SOP ***2*** : LAEO ***3*** : ML When set to a negative number, additional parameter settings can be added SOP: ***-1 L sl_cutoff dl_neigh rcut*** LAEO: ***-2 sigma k_cutoff k_neigh rcut*** ML: ***-3 alpha beta sigma L ml_cutoff ml_neigh rcut***
solid_cutoff	Float (0.8)	*MyTm* calculates the proportion of solid atoms according to the ***method*** specified, and when it is greater than ***solid_cutoff***, the system is judged as solid phase.
liquid_cutoff	Float (0.2)	*MyTm* calculates the proportion of solid atoms according to the ***method*** specified, and when it is less than ***liquid_cutoff***, the system is judged as liquid phase.

2. ensemble selection

md_script_interpreter	String	The interpreter name for script, such as python, bash, etc.
md_script_filename	String	The script path and name for MyTm to call MD.
md_steps	Integer (1000)	The number of steps in MD simulation.
md_steps_adaption	Integer (0)	Whether to automatically adjust the number of MD simulation steps. 0 represents No. 1 represents YES

3. temperature control

run	Command (None)	Temperature adjustment method.

		run heatingup startTem endTem heatRatio ***startTem*** and ***endTem*** defines the initial temperature control interval of ***heatingup, bisection*** and ***smallcell***. ***heatRatio*** defines the heating rate of ***heatingup.*** ***run bisection startTem endTem breakType threshold*** ***breakType*** equal to 1 defines the temperature convergence and ***breakType*** equal to 2 defines solid-liquid ratio stability accuracy of ***bisection.*** Parameter ***threshold*** defines convergence tolerance. ***run smallcell startTem endTem***
max_step	Integer (20)	Maximum number of iterations for temperature adjustment.
max_sub_step	Integer (40)	The maximum number of repeated calls to MD simulation in a single iteration.
bisection_relax_struct	Integer (0)	In the bisection method, the solid-liquid model is optimized for each step. 0 represents No. 1 represents YES
init_tem_check	Integer (1)	Check if ***startTem*** and ***endTem*** in the ***run*** command are reasonable. 0 represents No. 1 represents YES

4. model construction.

supercell	Integer (1 1 1)	The multiplier to expand the MD simulation box along the first lattice vector.
build	Command (None)	Constructing defects or coexisting solid-liquid phases. ***build type shape1 shape2 defect_scale*** ***type=defect*** or ***liquid.*** Create defect areas or liquid phase ***shape1/shap2*** **=0, 1, 2, 3**. Create geometric shapes for liquid

		phase regions or defect regions. ***shape1*** represents the interior of geometric shapes, while ***shape2*** represents the exterior. (0: None, 1: Parallelepiped shape, 2: Ellipsoid shape, 3: Cylinder shape) ***defect_scale*** represents the proportion of atoms removed from defects.
size	Command (None)	The size of the geometric region in the build command ***size a1 b1 c1 a2 b2 c2*** ***a1, b1, c1*** represent the size of the parallelepiped of ***shap1***. ***a2, b2, c2*** represent the size of the parallelepiped of ***shap2***.
pos	Command (None)	Center position of defect or liquid phase ***pos x1 y1 z1 x2 y2 z2***
rot	Command (None)	Rotation angle of defect or liquid phase along x, y and z axis. ***rot x1 y1 z1 x2 y2 z2***

C. How to build a script for MD calls

The main tasks of the MD interface script are as follows:

(1). Perform a MD simulation.

MyTm will write the model to be simulated by MD into *POSCAR_MyTm*, which is the POSCAR format of *VASP*. If it is not *VASP* running MD, format conversion is required, such as the ASE library in Python. (We have provided an example of the LAMMPS interface script in the *example* folder.)

The rule for calling scripts in *MyTm* is:

$(md_script_interpreter) $(md_script_filename) start_tem end_tem md_steps steps sub_steps

md_script_interpreter and ***md_script_filename*** are the script interpreters and script names set in the *input.dat* file. Therefore, the initial temperature, end temperature, MD simulation steps, *MyTm* iteration steps, and iteration sub steps of MD simulation are passed in through script parameters. As shown in Fig. 1, the temperature and number of steps for MD are obtained in lines 4-6 and set to *INCAR* in lines 13-15.

As shown in Fig. 1, after preparing the MD simulation model and setting the temperature and simulation steps, the MD software (*VASP*) is called in line 16.

(2). Post-processing after the MD simulation completes.

After calling the script once, *MyTm* will only read four files, namely *Temperature_MyTm*, *Pressure_MyTm*, *Volume_MyTm*, and *XDATCAR_MyTm.* These files are the temperature, pressure, volume, and trajectory during the MD simulation process. *Temperature_MyTm*, *Pressure_MyTm*, *Volume_MyTm*. The format of files *Temperature_MyTm*, *Pressure_MyTm* and *Volume_MyTm* is two columns, the first column is the simulated time steps, and the second column is the corresponding value. The *XDATCAR_MyTm* file is a VASP flavor trajectory file, which is a combination of multiple POSCAR files.

As shown in Fig. 1, lines 19-23 handle the four files mentioned above.

```
1   #!/bin/bash
2
3   # get parameters for MD
4   tem_start=$1
5   tem_end=$2
6   nsw=$3
7   steps=$4
8   sub_steps=$5
9
10  # run MD by VASP code
11  cp INCAR_init INCAR
12  cp POSCAR_MyTm POSCAR
13  sed -i "1i TEBEG=${tem_start}" ./INCAR
14  sed -i "1i TEEND=${tem_end}" ./INCAR
15  sed -i "1i NSW=${nsw}" ./INCAR
16  mpirun -n 32 vasp6.4_gam > vasp.log
17
18  #get MD trajectory to file XDATCAR_MyTm and temperature, pressure and vol
19  cp XDATCAR XDATCAR_MyTm
20  grep 'T=' OSZICAR |awk '{print NR,$3}' > Temperature_MyTm
21  grep 'volume of cell :' OUTCAR |awk '{print NR,$5}' > Volume_MyTm
22  sed -i '1,2d' Volume_MyTm
23  grep 'total pressure  =' OUTCAR |awk '{print NR,$4}' > Pressure_MyTm
24
25  #MD result backup
26  mv XDATCAR XDATCAR_${steps}_${sub_steps}
27  mv OUTCAR OUTCAR_${steps}_${sub_steps}
28  mv OSZICAR OSZICAR_${steps}_${sub_steps}
```

Figure 1. Example script file for *MyTm* calling external MD software (here VASP). Lines 3–8: Retrieve the MD simulation temperature, number of steps, and *MyTm* steps. Lines 10–16: Run the molecular dynamics simulation. Lines 18–23: Extract the MD simulation trajectories of temperature, pressure, and volume into corresponding files. Lines 26–28: Backup the MD simulation results.

(3). Other examples.

We provide script instances for LAMMPS and VASP in the example folder. The scripts for LAMMPS are processed using Python, with the only difference being that they are formatted using Python's ASE library.

D. Some precautions

1. The *input.dat* file generated automatically by *MyTm* cannot be directly used, and ***supercell***, ***run***, ***md_script_interpreter*** and ***md_script_filename*** commands need to be set according to the actual situation.
2. When using the smallcell method with the ***run*** command, the initial and end temperatures *MyTm* cannot be automatically adjusted, regardless of whether the ***init_tem_check*** command is set to 0 or 1.
3. When using ensembles without temperature control, such as NVE and NPH. It is strongly recommended to set a larger ***md_steps*** (such as 50000fs) and open ***md_steps_adaptation*** **.**
4. When using NVE or NVT ensembles, attention should be paid to pressure. The atomic density of the initial model usually needs to be adjusted to a density near the melting point.
5. Running errors usually come from two aspects: the MD simulation did not end correctly and the results of the MD simulation were not correctly extracted into the *Temperature_MyTm, Pressure_MyTm, Volume_MyTm, XDATCAR_MyTm* files.
6. The melting point temperature of the Z method should be the end temperature of the last liquid phase. And *MyTm_result.dat* usually outputs the temperature of the solid phase. Use 'grep IF MyTm_result.dat' to extract information for each step.